

SHUBHIKA PRADEEP

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B.Tech – Computer Science (AI & ML), PES University (2024–2028)

TECHNICAL SKILLS

Languages: Python, C, JavaScript, R, SQL

ML/AI: PyTorch, TensorFlow, scikit-learn, NLP, LLM/SLM systems, Recommendation Systems

Web: React, Flask, Express, REST APIs, Svelte, HTML/CSS

Data/Tools: Pandas, NumPy, Git, MySQL, MongoDB, Botpress, Zapier, Qiskit

EXPERIENCE

Research Scholar — QuaNAD Lab, PES University | 2024–Present

- Built ML models (CNN, MLP, KNN) to classify topological phases using Hamiltonian-based representations
- Implemented physics-informed data generators (Fourier + SSH model) with winding number computation
- Achieved high accuracy on clean datasets; analyzed **generalization failure** on unseen winding numbers
- Evaluated robustness under **distribution shifts and parameter variations**
- Introduced **disorder (noise in hopping parameters)** to study phase transitions and model breakdown
- Constructed and analyzed **phase diagrams** to compare ML predictions vs physical ground truth
- Demonstrated that models rely on statistical patterns rather than true topological invariants

Teaching Lead & Research Contributor — Stanford Code in Place | 2024, 25, 26

- Selected as Teaching Lead for Stanford's global Python education initiative.
- Mentored international cohorts in Python, debugging, and algorithmic thinking.
- Contributed to SLM-based research for automated student feedback evaluation.

AI Automation Intern — Taskformer (Dallas Mental Wellness Centre) | 2024

- Built and deployed AI chatbots and workflow automations using Python, Botpress, and Zapier.

- Implemented intent classification, safety checks, and ethical response handling for mental health use cases.
 - Integrated conversational systems into real-world healthcare operations.
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PROJECTS

Mappeer – ML-Based Sustainable Commute App | Python, ML, Full-Stack

- Developed a commute-matching platform with ML-powered route recommendations.
- Modeled user behavior to optimize eco-friendly transport choices.
- Integrated ML into a deployable full-stack application.

Mental Health Diagnostic System | C, Systems Programming

- Implemented PHQ-9 and GAD-7 diagnostic assessments in C with CSV-based processing.
- Built scoring logic, structured data handling, and report generation.
- Applied low-level systems design to healthcare analytics.

ADDfocus – ADHD Productivity Tool | Cognitive UX

- Built a cognition-aware productivity system with task structuring and timed workflows.
- Designed features to reduce overload and improve focus for neurodivergent users.

Stock Market Prediction System (R) | Neural Networks, Time-Series

- Built regression and neural-network models for stock trend forecasting.
 - Performed preprocessing, feature engineering, and evaluation on historical data.
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LEADERSHIP & ACTIVITIES

- **CTF Question Setter — Layer8 & IEEE:** Designed exploit-based challenges for learning-focused hackathons; Top 20 nationally in CTF competition.
 - **Technical/Operations Teams — IEEE, Neural Hive, Layer8, Maaya:** Supported hackathons, websites, tooling, and large-scale event logistics.
 - Club Head - WEAL, The flagship Mental Health Club of PESU
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CERTIFICATIONS

Harvard ManageMentor | Stanford Code in Place Teaching Lead Training | Microsoft Learn Student Ambassador (MLSA)